

McLean, VA | +1 315-952-4519 | rushiashokbhaisavani@gmail.com | <https://linkedin.com/in/rushisavani/> | [GitHub](#) | [Portfolio](#)

AI/ML Engineer with 3+ years of experience building and deploying Generative AI, Agentic AI, NLP, and production ML solutions across healthcare, enterprise, and retail environments. Expertise in LLMs, RAG, text-to-SQL, LangGraph, LLM inference optimization (vLLM, CUDA, quantization), PyTorch, MLflow, Docker, Kubernetes, and LLMOps, with a strong focus on scalable, reliable, and cost-efficient AI systems. Delivered measurable business impact including 90% faster data lookups, 24% fewer AI production issues, and \$1M+ in annual savings.

Languages & Development: Python, SQL, TypeScript, Next.js, Git, REST APIs, FastAPI, Postman, Jupyter

Machine Learning: Scikit-learn, XGBoost, Gradient Boosting, SVM, Decision Trees, HDBSCAN, Feature Engineering, Hyperparameter Tuning, Cross-Validation, Classification, Regression, A/B Testing, Hypothesis Testing

Deep Learning & NLP: PyTorch, CUDA, TensorFlow, Hugging Face Transformers, BERT, RoBERTa, Sentence-BERT, Embeddings, Fine-Tuning, Quantization (FP16/INT8/INT4), Sentiment Analysis

Generative AI & LLM: LLMs, RAG, Agentic AI, Text-to-SQL, Model Context Protocol (MCP), LangChain, LangGraph, LlamaIndex, Google ADK, LiteLLM, vLLM, LLM Inference Optimization, KV Caching, Continuous Batching, Prompt Engineering, Guardrails, RAGAS, LangSmith, LLMOps

MLOps & Cloud: AWS, Microsoft Azure, GCP, SageMaker, EKS, GKE, OpenSearch, DynamoDB, Redshift, CloudWatch, MLflow, Docker, Kubernetes, Horizontal Pod Autoscaler (HPA), GitHub Actions, CI/CD, Redis, OpenTelemetry, Model Monitoring

Data & BI: PySpark, Pandas, NumPy, SQL Server, Supabase, Snowflake, PostGIS, pgvector, Qdrant, ETL/ELT, Data Modeling, Power BI, Tableau

Associate Researcher | Syracuse University

Sep 2026 – Present

- Conducted data analysis and research using Python, SQL, and statistical techniques to analyze complex datasets, identify meaningful trends, and generate actionable insights to support academic research initiatives.
- Ran Hypothesis testing to identify significant variables for target outcomes, plan feature selection, model choice and validation steps for training and evaluating ML models.

Founder, Software Engineer | [ParkOne](#) | Remote, India | [Instagram](#)

Jan 2026 – Sep 2026

- Grew ParkOne from zero to 200+ unique users across 3 Mumbai neighborhoods as the sole engineer, shipping proximity search with PostGIS, Mapbox maps, and Google Geocoding on a Next.js and TypeScript stack.
- Designed the Supabase PostgreSQL schema, auth, and a PostGIS radius-search function behind spot search, matching active parking segments within a user-set radius (default 500m) across each segment's start, midpoint, and end points.
- Interviewed 50+ drivers and 30+ parking owners to validate demand, which shifted the product from showing only where parking is possible to detailed parking-area information with a clearer map view, while handling early go-to-market alone.

AI Engineer / Data Science | ProHealth Home Health and Hospice | Atlanta, GA

Jun 2025 – May 2026

- Built an agentic text-to-SQL pipeline on AWS using Python, LangGraph, MCP tooling, document knowledge retrieval, DynamoDB for session memory and self-correcting loops, cutting administrator lookup time by 90%.
- Diagnosed inconsistent retrieval on a 100+ question eval set using RAGAS and human-in-the-loop review, fixing with OpenSearch hybrid search and reranking, reaching 93% context precision and 98% faithfulness.
- Engineered an ETL pipeline using Python, Apache Spark and SQL to ingest, transform and consolidate patient, financial and operational data into 30+ reusable views, replacing manual processing time by 45%.
- Retrained a readmission ML model on AWS SageMaker using Python, scikit-learn and clinical feature engineering, running 30+ experiments through MLflow and cutting readmission cases by 10% (~\$15K savings).
- Architected three LangChain pipelines with parameterized SQL and deterministic chunking across predefined data points, giving finance and admin teams reproducible, auditable root-cause reports using LLMs, saving \$80K annually.
- Reduced AI production issues 24% by tracking token usage, cost-per query, and hallucination rates live in OpenTelemetry and LangSmith, backed by automated evals and HIPAA-compliant guardrails.
- Deployed AI Inference services on AWS using Docker and Kubernetes, automating releases with GitHub Actions CI/CD and cutting model-serving costs by 17% via Redis caching and routing optimization.

Data Science Research Assistant | Syracuse University | Syracuse, NY

Jan 2025 – May 2025

- Fine-tuned a RoBERTa-based sentiment classification model using PyTorch and Hugging Face Transformers on 7,000+ survey responses, applying class-weight optimization and stratified k-fold validation to improve labeling accuracy by 30%.
- Applied Sentence-BERT embeddings and HDBSCAN clustering to identify distinct AI-adoption behavior groups from survey data.
- Analyzed sentiment, AI-adoption patterns, and KPIs to derive actionable research insights.
- Developed Power BI visualizations and analytical summaries to communicate sentiment distributions, behavioral segments, and KPI trends to research stakeholders.

Data Scientist | IBM | Mumbai, India

Jan 2023 – Jun 2024

- Mitigated \$1.2M in operating expense by lifting baseline precision by 18% with tree-based (XGBoost, Random Forest) and logistic regression models, on Azure ML for risk scoring and demand forecasting.
- Improved downstream data ingestion latency 35% by automating ETL and feature engineering pipelines across Azure Data Factory and SQL for 1M+ daily transaction records.
- Optimized reporting data pipeline using star schema and refactored stored procedures on Azure PostgreSQL, bringing query time down 40% and enabling sub-second dashboards for multiple clients.
- Spearheaded rigorous statistical model validation, improving model performance through Bayesian hyperparameter tuning using Optuna, eliminating data leakage and accelerating model selection cycle by 20%.
- Scoped loosely defined business questions from product managers and BI teams into defined model outputs and KPIs, presenting through Tableau dashboards, eliminating ad hoc reporting work by 30%.

Distributed LLM Inference Platform | Python, vLLM, PyTorch, GCP, Kubernetes

- Optimized a PyTorch CUDA engine on NVIDIA L4 (naive, KV cache, continuous batching, vLLM) at 214 vs 34 tok/s at concurrency 16, cut O(n^2) decode to O(n) (~5x), FP16/INT8/INT4 (980 to 489 MiB), GKE HPA 1 to 4 pods. | [GitHub](#)

PrivateRAG | Python, Qdrant, Llama 3.1, TypeScript, LlamaIndex

- Designed a fully local RAG system for private-document Q&A with Qdrant vector indexing, metadata filtering, and Llama 3.1 inference, featuring source attribution and chunk-level traceability. | [GitHub](#)

AI Email Response System | Python, RAG, MCP, LLM-as-a-Judge

- Implemented an automated RAG-based email response workflow with policy checks, MCP-based routing, LLM-as-a-Judge validation, and human handoff for high-risk cases, reducing manual investigation by 70%. | [GitHub](#)

M.S. Information Systems (Data Science), Syracuse University — GPA: 3.86/4.00

Aug 2024 – May 2026

B.E. Information Technology, Thadomal Shahani Engineering College, Mumbai — GPA: 8.52/10

Aug 2020 – Jun 2024